Answer Relevance, Worked Out by Hand

The metric that catches evasive, padded, or off-topic answers — it reverse-engineers questions *from* the answer and measures how close they are to the one actually asked, using cosine similarity of embeddings.

What this document does. It defines answer relevance via a clever trick — ask an LLM to generate the questions a given answer would perfectly address, embed them, and compare to the original question — then computes the score as the mean cosine similarity. It explains why this catches dodging and verbosity that faithfulness misses. Numbers from Appendix A.

One of four RAGAS metrics: it pairs with faithfulness (#25). This one asks "did you answer *the question*?"; faithfulness asks "is the answer *grounded*?"

1 What it measures

An answer can be perfectly faithful (every claim grounded) and still useless — vague, padded, or quietly changing the subject. Answer relevance scores how well the response *actually addresses the question asked*, independent of whether its facts are correct.

Intuition

The trick is to run the model backwards. If an answer genuinely addresses a question, then someone reading *only the answer* should be able to guess what was asked. So: hand the answer to an LLM and say "what questions would this answer perfectly respond to?" If those reverse-engineered questions look like the real one, the answer was on-target. If the answer was evasive or rambling, the guessed questions drift away from the original — and the similarity drops. Relevance becomes a measurable distance.

2 The procedure and formula

1. From the generated answer, prompt an LLM to produce n candidate questions $q'_1, \dots, q'_n$ it would answer.
2. Embed the original question q and each generated q'_i (Track 1).
3. Average the cosine similarities:

$$\boxed{\text{Answer Relevance} = \frac{1}{n}\sum_{i=1}^{n} \cos\big(\text{emb}(q),\, \text{emb}(q'_i)\big)} \in [-1, 1] \text{ (usually } [0,1]).$$

3 Worked by hand

Original question. "How tall is the Eiffel Tower?"
Generated answer. "The Eiffel Tower stands 330 metres tall."
Prompt an LLM to reverse-engineer $n = 3$ questions this answer fits, embed them, and cosine-

compare to the original:

reverse-generated question q_i'	$\cos(\mathrm{emb}(q), \mathrm{emb}(q_i'))$
"What is the height of the Eiffel Tower?"	0.91
"How many metres tall is the Eiffel Tower?"	0.84
"What is the Eiffel Tower's height in metres?"	0.88

$$\text{Answer Relevance} = \frac{0.91 + 0.84 + 0.88}{3} = \frac{2.63}{3} = \boxed{0.8767}.$$

All three reconstructed questions are paraphrases of "how tall" — the answer clearly targeted the question, so the score is high.

Mini-example · how dodging tanks the score

Suppose the answer were instead "The Eiffel Tower is a famous landmark in Paris visited by millions." An LLM reading that would reverse-generate questions like "Where is the Eiffel Tower?" or "Why is the Eiffel Tower famous?" — whose embeddings sit *far* from "How tall is it?", pushing cosines toward 0.3–0.5 and the mean well down. The answer is perfectly *faithful* (those facts are true and grounded) yet *irrelevant* to what was asked — and answer relevance is the only one of the four metrics that catches it.

4 Why generate questions instead of comparing answers?

A simpler idea — embed the question and the answer and take their cosine — fails, because a question and its answer are phrased very differently ("How tall...?" vs "330 metres") and naturally have modest similarity even when perfectly matched. By mapping the answer *back into question-space*, both sides are now questions and the comparison is apples-to-apples. The detour through generated questions is what makes the cosine meaningful.

Intuition

Two failure signatures this metric exposes, both invisible to faithfulness: **evasion** (the answer addresses a *different*, easier question — reverse-questions drift off topic) and **verbosity** (the answer buries the response in padding — reverse-questions become scattered and unfocused, lowering average similarity). High relevance demands a *focused, on-topic* answer, not just a true one.

Equation cheat-sheet

$$\text{Answer Relevance} = \frac{1}{n}\sum_{i=1}^{n} \cos\left(\mathrm{emb}(q), \mathrm{emb}(q_i')\right)$$

q_i' = questions an LLM reverse-generates from the answer; catches evasion & padding

Appendix A · Reference sketch

```
import numpy as np
# q'_i = LLM-generated questions from the answer; cosines vs the original question
sims = [0.91, 0.84, 0.88]
answer_relevance = float(np.mean(sims))      # 0.8767

# A dodging answer would yield off-topic q'_i with low cosines, e.g.
# [0.42, 0.31, 0.50] -> 0.41  (faithful but irrelevant)
```